WASTE HEAT TO ELECTRICAL ENERGY CONVERSION USING THERMOELECTRIC GENERATOR

With Smart Power Management

Turning Waste Heat into a Sustainable Future

BY
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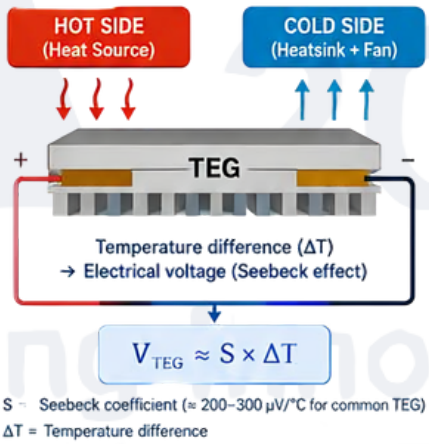
• Project Idea:

This project is based on the idea of converting waste heat to electrical energy. A significant amount of energy is continuously lost as waste heat from automobile exhaust systems, industrial machinery, cooking systems, generators, and other thermal processes. This project proposes a compact Waste Heat to Electrical Energy Conversion System using a Thermoelectric Generator (TEG) to convert a temperature difference directly into electrical energy through the Seebeck effect.

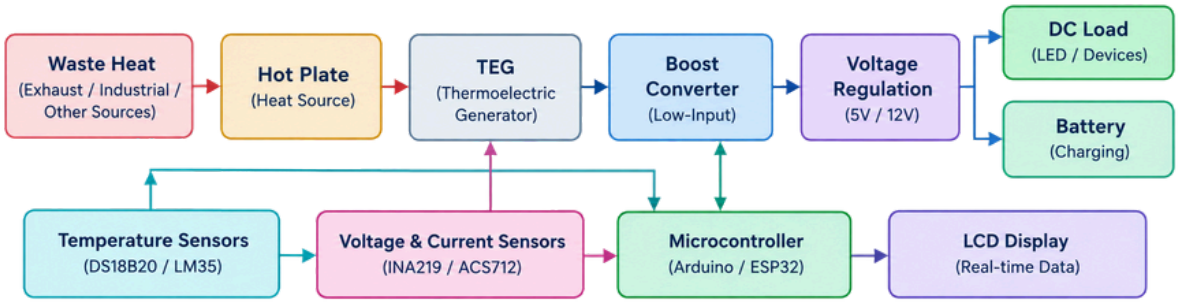
The system consists of a TEG placed between a hot surface and a cooled heatsink. The generated low DC voltage is boosted to a useful level using a low-input-voltage DC-DC boost converter, then regulated to supply a low-power load and charge a rechargeable battery through a suitable charging circuit. Temperature, voltage and current are measured, and an optional microcontroller displays real-time data such as temperatures , △T, voltage, current, power and efficiency.

This project demonstrates a low-maintenance, compact and environmentally friendly method of recovering useful electrical energy from waste heat, with potential applications in remote sensors, industrial monitoring. exhaust-heat recovery and other areas where a temperature gradient is available.

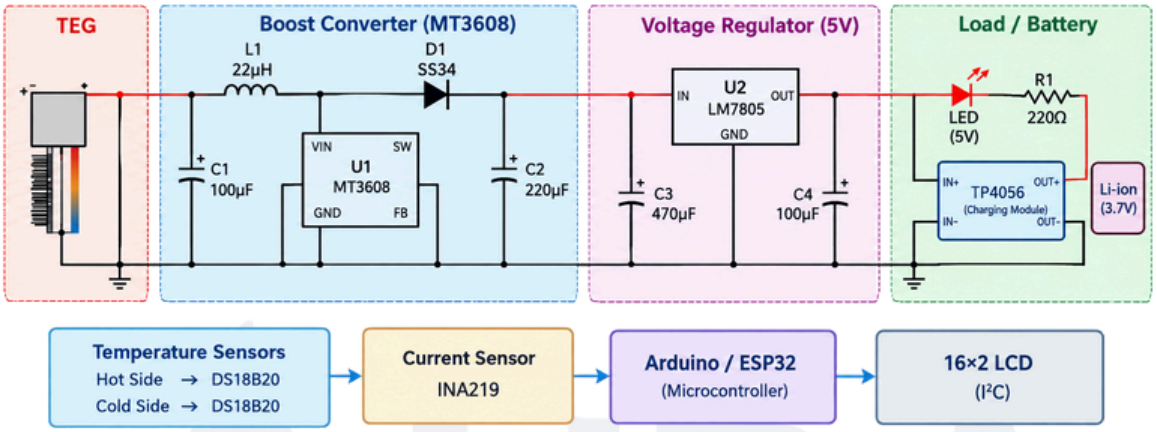
• Working Principle:



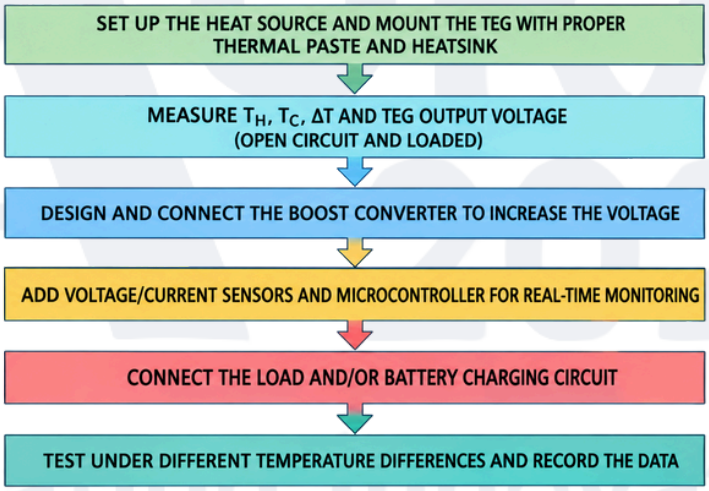
• Block Diagram:



- **Circuit Diagram:**

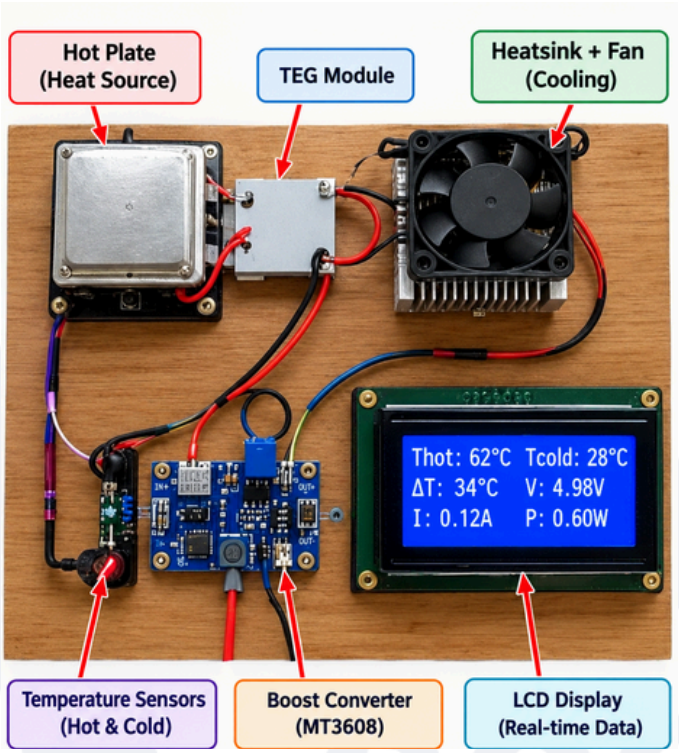


- **Methodology:**



- **Key Features:**
  - Converts waste heat to useful electrical energy
  - Boost converter for low input voltage
  - Real – time monitoring (temperature, voltage, current, power, efficiency)
  - Battery charging capability
  - Compact and low maintenance
  - Ecofriendly and sustainable solution

- **Experimental Set-Up:**



- **Expected Results:**

Parameter	Expected Value (Example)
Hot side temperature (T_H)	60 – 90 °C
Cold side temperature (T_C)	25 – 35 °C
ΔT	25 – 60 °C
TEG output voltage (V_{in})	0.8 – 2.5 V
Boosted output voltage (V_{out})	5 V
Output current (I)	0.05 – 0.2 A
Output power (P = V × I)	0.25 – 1.0 W
System efficiency (estimated)	5 – 12%

  - Values will vary based on the TEG module, heat source and operating conditions.

- **Applications:**



- **Automobile exhaust heat recovery**



- **Industrial Machinery**



- **Cooking systems and stove**



- **Generators and power plants**



- **Remote sensors and IoT devices**



- **Off-grid and nural electrification**

- **Conclusion:**

The project demonstrates the practical use of TEG and power electronics to convert wasted heat energy to useful electrical energy. It provides a sustainable , low maintenance and eco-friendly solution and various applications in different sectors.

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